YOJASREE PALICHERLA

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PROFESSIONAL SUMMARY

Data Analyst and IT graduate student with over one year of hands-on experience in data cleaning, transformation, and exploratory analysis using SQL, Python, and Excel. Developed interactive Tableau dashboards and Power BI reports to merge and visualize complex datasets for actionable insights. Hands-on experience building LLM-powered applications, including retrieval-augmented generation (RAG) pipelines, vector search, and prompt-engineered solutions. Proven track record collaborating with diverse teams and communicating findings to support decision-making.

CORE EXPERTISE

Machine Learning: Supervised & Unsupervised Learning, Deep Learning, Model Evaluation, Data Preprocessing
LLMs & Generative AI: LLM Experimentation, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Search & Embeddings, Semantic Retrieval
Deep Learning & NLP: PyTorch, Hugging Face Transformers, Transfer Learning, Fine-Tuning Pretrained Models (DistilBERT)
Programming & Development: Python, SQL, Git/GitHub
Tools & Frameworks: Scikit-learn, Pandas, NumPy, SciPy, Tableau, Power BI, MS Excel, SAS, SPSS, FAISS/Chroma (Vector Databases), Microsoft Azure, GitHub Copilot
Cloud & Big Data: Microsoft Azure; familiarity with cloud-based analytics platforms (Snowflake, Databricks); experience handling large-scale datasets (50K+ rows)
Industry Experience: Healthcare / Health Insurance analytics (PwC)
Data Management & Reporting: Data Cleaning, Feature Engineering, Model Optimization, Data Reporting, Data Auditing, Data Visualization
Other Skills: Problem Solving, Critical Thinking, Team Collaboration, Interpersonal & Leadership Skills, Technical Documentation

EDUCATION

University of Potomac — Master of Science, Information Technology — **Jan 2026 - Present**
Sri Vidyaniketan Degree College — Bachelor of Science, Computer Science — **Jul 2024**

PROFESSIONAL EXPERIENCE

SmartInternz — **Oct 2023 - Apr 2024**
Data Analytics

- Completed a virtual "Data Analytics with Tableau" internship focused on end-to-end data analysis, visualization, and reporting.
- Performed data cleaning, transformation, and exploratory analysis using Excel, SQL, and Python; developed interactive dashboards to present key insights.
- Partnered with mentors to interpret trends and anomalies in large datasets, strengthening data storytelling and stakeholder communication skills.
- Conducted ad hoc analyses and data reporting to address business questions and validate findings.
- Presented analytical reports and visualizations, strengthening skills in storytelling with data and stakeholder communication.
- Built calculated fields, filters, and parameter controls in Tableau to create dynamic, reusable dashboard templates for recurring reporting needs.

PwC Hyderabad — **Jul 2024 - Sep 2025**
AI & Data Analyst

- Conducted research on Generative AI tools and explored practical applications in business workflows, including early experimentation with LLM prompting techniques.
- Contributed to healthcare-focused AI initiatives, applying data analysis and AI-driven insights to support healthcare-related business use cases.
- Utilized Microsoft Azure to support data processing and AI-related workflows.
- Collaborated with cross-functional teams to document data findings and model experimentation results.
- Created interactive reports using Power BI to enable stakeholders to explore business metrics and enhance data visualization.
- Built and maintained SQL queries to extract and validate data from multiple sources, ensuring accuracy for downstream analysis.
- Communicated technical findings to non-technical stakeholders through clear visualizations and concise written summaries.
- Utilized SPSS and SAS to conduct empirical research and answer questions with data.

PROJECTS

AI-Powered Document Q&A System (RAG) — GitHub

- Designed and built a Retrieval-Augmented Generation pipeline to answer questions over a document corpus, reducing manual lookup time.
- Implemented text chunking and embedding generation (OpenAI/sentence-transformers) with a vector database (FAISS/Chroma/Pinecone) for semantic retrieval.
- Built a lightweight evaluation harness to measure answer accuracy against ground-truth answers, achieving 100% accuracy on a test set and reducing hallucinated responses through refined retrieval and prompt design.
- Deployed an interactive demo (Streamlit) allowing users to query documents in natural language with sourced, grounded answers.

Sentiment Classification with Fine-Tuned DistilBERT (NLP)

- Fine-tuned a pretrained DistilBERT transformer model (PyTorch, Hugging Face) for binary sentiment classification on the IMDb dataset (50,000 reviews), achieving 93–95% accuracy.
- Preprocessed and tokenized text data using Hugging Face Datasets and Tokenizers, applying padding and truncation to handle variable-length reviews.
- Leveraged transfer learning and hyperparameter tuning to reduce training time and improve performance over training from scratch.
- Evaluated model performance using precision, recall, and F1-score alongside accuracy to ensure balanced classification across both sentiment classes.

Customer/Sales Data Analysis Dashboard

- Cleaned and audited a 50K+ row e-commerce transactions dataset using Python and SQL, improving data reliability by 20%.
- Conducted empirical analysis revealing a 15% higher churn rate among paid-ad customers vs. organic, informing targeted retention strategies.
- Developed an interactive Tableau dashboard for non-technical stakeholders, reducing time-to-insight from hours to minutes.
- Collaborated with cross-functional teams to define data requirements and deliver regular compliance and performance reports.
- Applied segmentation techniques to group customers by behavior and value, supporting more targeted marketing and retention decisions.